

Equipment Rental & Booking System

Objective

To develop a web application that allows users to rent equipment online, check availability, make bookings, and return equipment. Administrators can manage equipment, bookings, and users through a secure dashboard.

Technology Stack

- **Frontend:** React.js
- **Backend:** Spring Boot
- **Database:** PostgreSQL
- **Authentication:** Spring Security + JWT
- **API Testing:** Postman

Modules

1. User Authentication

- User Registration
- User Login
- JWT Authentication

2. Equipment Management

- Add Equipment
- Update Equipment
- Delete Equipment
- View Equipment

3. Booking Module

- Book Equipment
- View Bookings
- Cancel Booking

4. User Dashboard

- View Profile
- Booking History
- Active Rentals

5. Admin Dashboard

- Manage Users
- Manage Equipment
- View All Bookings

External APIs

1. OpenStreetMap API

- Display rental shop location.

2. Email API (Brevo/EmailJS)

- Send booking confirmation and return reminders.

Database Tables

- Users
- Equipment
- Bookings